MERRA21C-ML: GENERATIVE CONUS DOWNSCALING

Native GEOS-FP (~25 km) → variable-specific conditional flow matching → six fields on the HWT (~3 km) grid

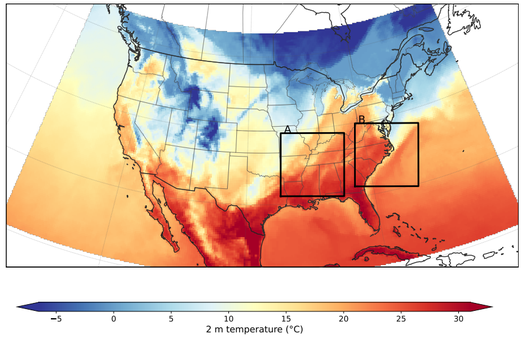
GEOS-FP + high-resolution GEOS replay + v4.1 predictions | 2026-03-07 19:30:00 UTC | epoch 243, EMA, 8 members, spectral

A COARSE WEATHER INPUTS

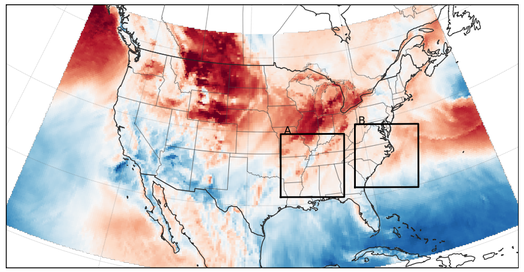
Current + previous hour; static and time features

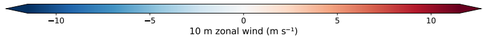
Native 0.25° × 0.3125° cells

2 m temperature



10 m wind: zonal component





Six weather fields at t − 1 h and t

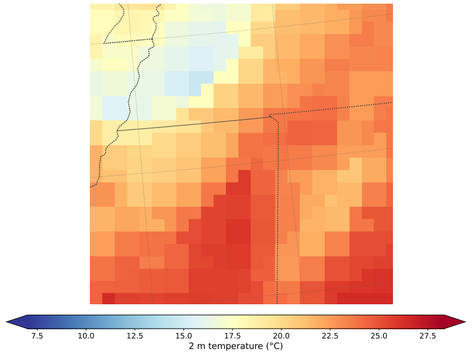
Local 192² patches + broad 576² context

Terrain, land/sea, coordinates, cell area, time

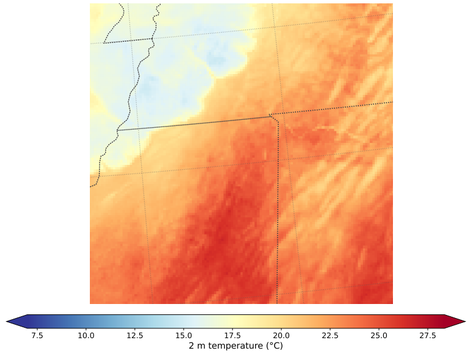
B SAME REGION, TWO GRIDS

3× closer: identical extent and scale; no interpolation

~25 km: native cells



~3 km: GEOS replay reference (HWT)



Temporal pairing and training-only statistics

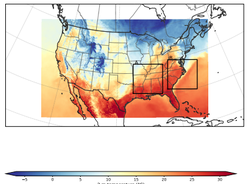
States: midpoint | Rain: ¼ P_{-30} + ½ P_0 + ¼ P_{+30}

Chronological splits with gaps; packed training data

C SIX JOINT OUTPUT FIELDS

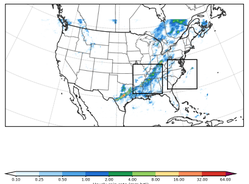
Actual v4.1 model member on the high-resolution GEOS grid

2 m temperature



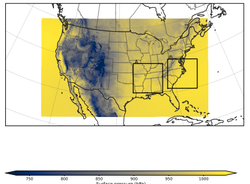


Hourly mean rain rate



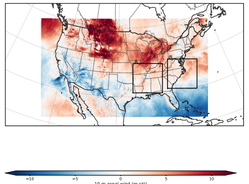


Surface pressure



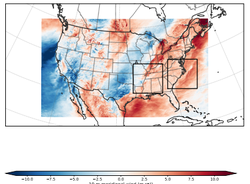


10 m zonal wind



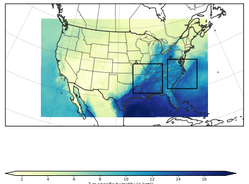


10 m meridional wind





2 m specific humidity





Variable-specific generative coordinates

Rain: direct $\sqrt{(1 + P/P_0)} - 1$; P_0 = 1 mm h⁻¹

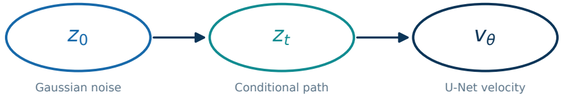
T, pressure, winds: regression residuals

Humidity: coarse-field residual

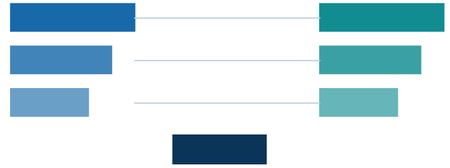
D REGRESSION → GENERATION

Frozen conditioner + local and broad-context U-Net

CONDITIONAL FLOW MATCHING



$z_t = (1 - t)z_0 + tz_1$ | target: $z_1 - z_0$



Residual blocks + broad-context conditioning

AREA- AND CHANNEL-WEIGHTED VELOCITY LOSS



192² input = 128² core + 32-pixel halo

Loss supervises core and halo

SYNCHRONIZED TILED ENSEMBLE SAMPLING

Blend tile velocities → shared-state Heun update

Independent seeds → multiple conditional members



One coarse input → 8 generated high-resolution members

Raw / spectral: paired members, no retraining

Maps: native GEOS-FP, HWT dynamical GEOS replay reference and actual v4.1 predictions | 2026-03-07 19:30:00 UTC | 64 Heun steps, spectral.

Temperature panels show the same 3× tighter crop with unchanged source cells and color scale; flow matching learns downscaling to the GEOS reference grid.